LAB EXPERIMENTS 15/07/2026

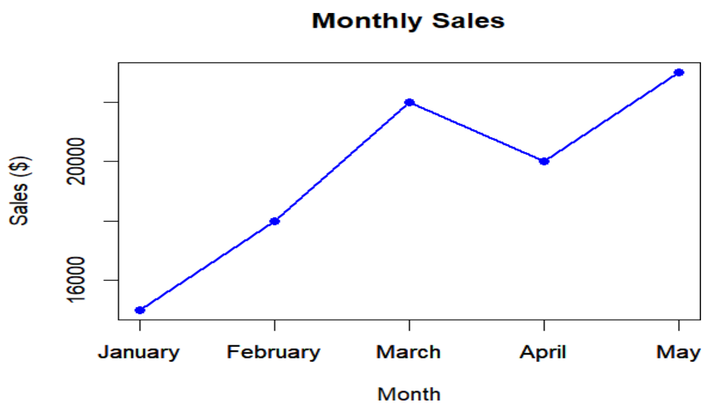
Set-01: Monthly Sales Data

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

Program:

```
sales_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
print(sales_data)
plot(
sales_data$Sales,
type = "o",
  col = "blue",
  pch = 16,
  lwd = 2,
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales"
)
axis(1, at = 1:5, labels = sales_data$Month)
```

Output:

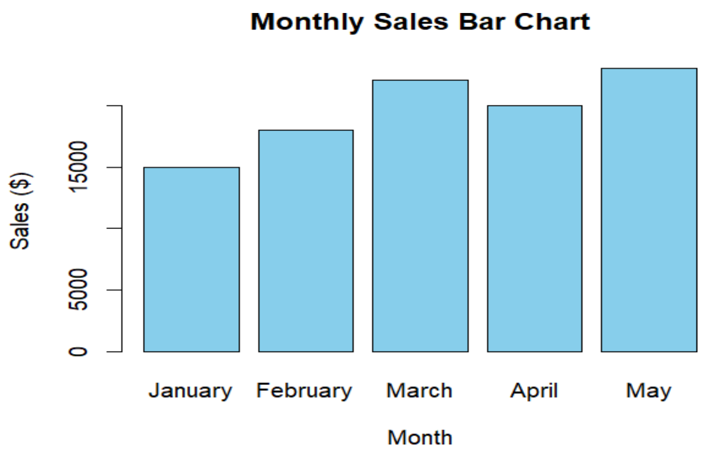


2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

Program:

```
# Create Dataset
sales_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
# Bar Chart
barplot(
sales_data$Sales,
  names.arg = sales_data$Month,
  col = "skyblue",
  border = "black",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Monthly Sales Bar Chart"
)
```

Output:

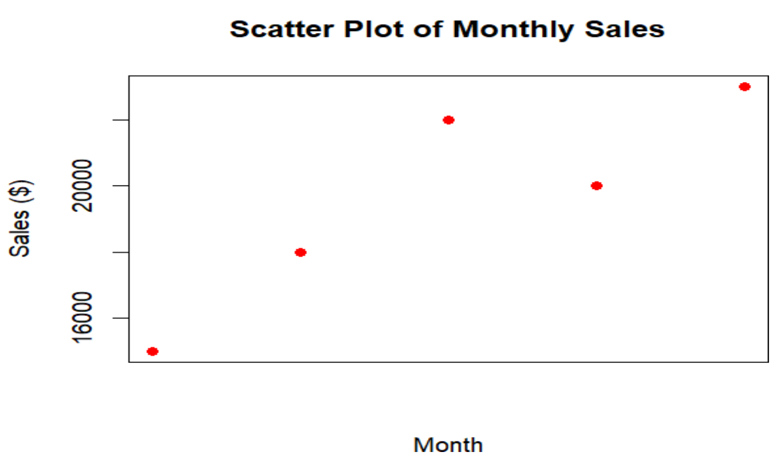


3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

Program:

```
# Create Dataset
sales_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
# Scatter Plot
plot(
  x = 1:5,
  y = sales_data$Sales,
  pch = 19,
  col = "red",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales ($)",
  main = "Scatter Plot of Monthly Sales"
)
# Display month names on x-axis
axis(1, at = 1:5, labels = sales_data$Month)
```

Output:



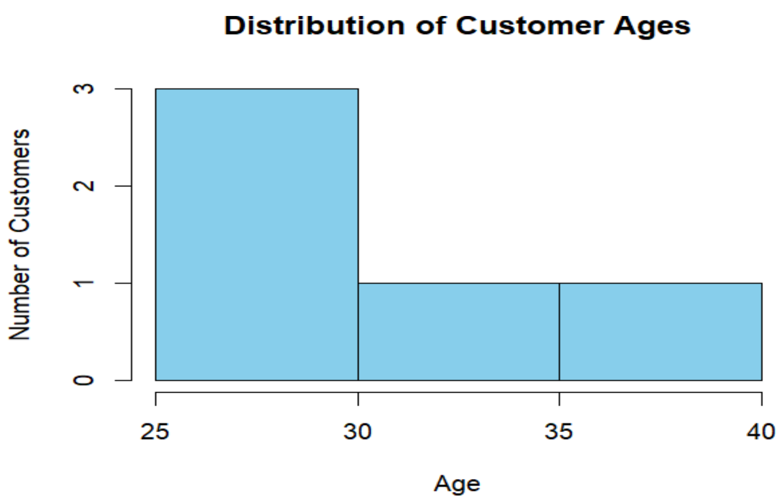
Set-02: Customer Feedback Analysis

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

Program:

```
customer_data <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  SatisfactionScore = c(4, 5, 3, 4, 5)
)
customer_data
hist(
  customer_data$Age,
  breaks = 5,
  col = "skyblue",
  border = "black",
  main = "Distribution of Customer Ages",
  xlab = "Age",
  ylab = "Number of Customers"
)
```

Output:

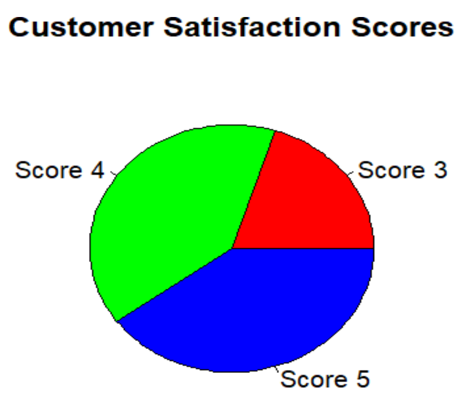


2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

Program:

```
customer_data <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
  Age = c(25, 30, 35, 28, 40),
  SatisfactionScore = c(4, 5, 3, 4, 5)
)
score_count <- table(customer_data$SatisfactionScore)
pie(
  score_count,
  labels = paste("Score", names(score_count)),
  col = rainbow(length(score_count)),
  main = "Customer Satisfaction Scores"
)
```

Output:



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

Program:

```
customer_data <- data.frame(
  CustomerID = c(1, 2, 3, 4, 5),
```

```
  Age = c(25, 30, 35, 28, 40),
  SatisfactionScore = c(4, 5, 3, 4, 5)
)
customer_data$AgeGroup <- cut(
  customer_data$Age,
  breaks = c(20, 30, 40, 50),
  labels = c("21-30", "31-40", "41-50"),
  include.lowest = TRUE
)
age_table <- table(
  customer_data$AgeGroup,
  customer_data$SatisfactionScore
)
print(age_table)
barplot(
t(age_table),
  col = c("red", "orange", "green"),
  legend.text = TRUE,
  xlab = "Age Group",
  ylab = "Number of Customers",
  main = "Customer Satisfaction by Age Group"
)
```

Output:

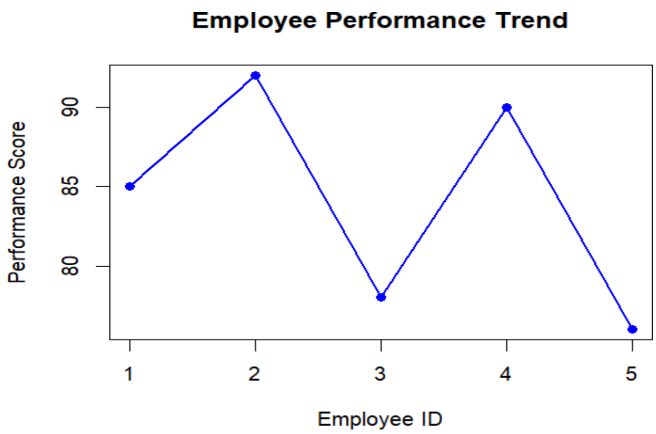
Set-03: Employee Performance Evaluation

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Program:

```
# Employee Performance Dataset
employee_data <- data.frame(
EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsOfService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)
# Line Chart
plot(
  employee_data$EmployeeID,
  employee_data$PerformanceScore,
  type = "o",
  col = "blue",
  pch = 16,
  lwd = 2,
  xlab = "Employee ID",
  ylab = "Performance Score",
  main = "Employee Performance Trend"
)
legend(
  "bottomright",
  legend = "Performance Score",
  col = "blue",
  lty = 1,
  pch = 16
)
```

Output:

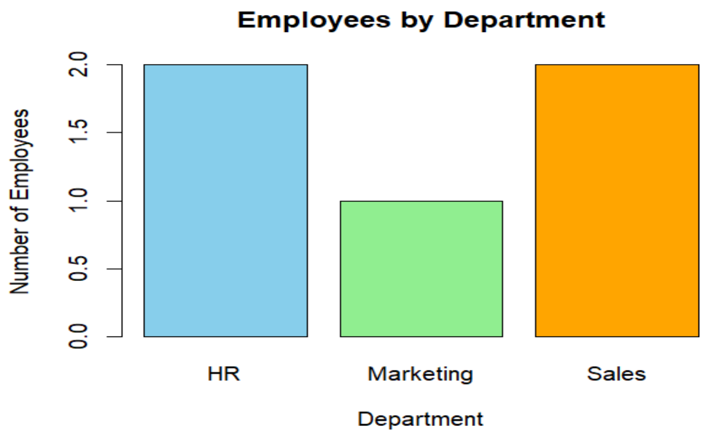


2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Program:

```
# Employee Performance Dataset
employee_data <- data.frame(
EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsOfService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)
# Department Count
dept_count <- table(employee_data$Department)
# Bar Chart
barplot(
  dept_count,
  col = c("skyblue","lightgreen","orange"),
  border = "black",
  main = "Employees by Department",
  xlab = "Department",
  ylab = "Number of Employees"
)
```

Output:



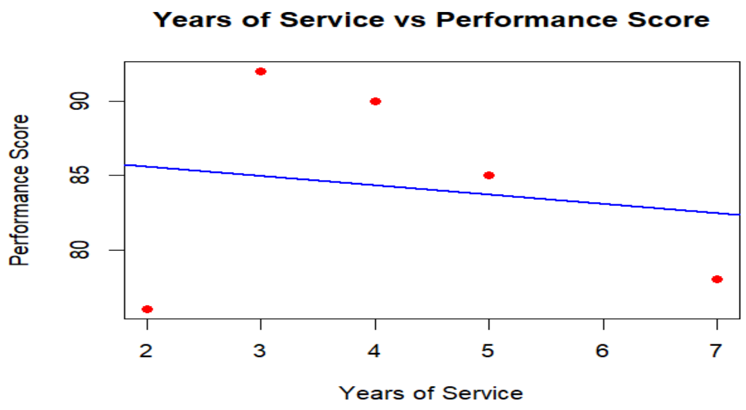
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

Program:

```
# Employee Performance Dataset
employee_data <- data.frame(
EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsOfService = c(5,3,7,4,2),
  PerformanceScore = c(85,92,78,90,76)
)
# Scatter Plot
plot(
  employee_data$YearsOfService,
  employee_data$PerformanceScore,
  pch = 19,
  col = "red",
  xlab = "Years of Service",
  ylab = "Performance Score",
  main = "Years of Service vs Performance Score"
)
# Trend Line
```

```
abline(
  lm(PerformanceScore ~ YearsOfService, data = employee_data),
  col = "blue",
  lwd = 2
)
```

Output:



Set-04: Product Inventory Management

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Program:

```
# Load library
library(ggplot2)
# Create dataset
inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220)
)
# Bar Chart
ggplot(inventory, aes(x = Product_Name,
                      y = Quantity_Available,
                      fill = Product_Name)) +
  geom_bar(stat = "identity") +
```

```
  labs(
    title = "Quantity Available for Each Product",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

Output:

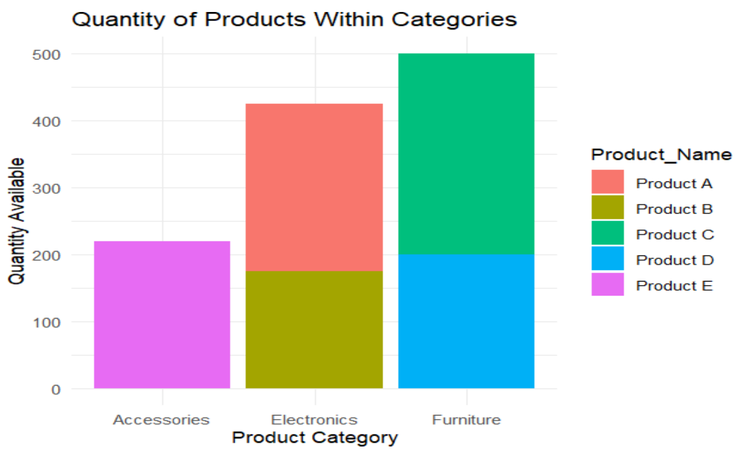
2. Generate a stacked bar chart to show the quantity of each product within different product categories.

Program:

```
# Load library
library(ggplot2)
# Create dataset
inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity_Available = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Accessories")
)
# Stacked Bar Chart
ggplot(inventory, aes(x = Category,
                      y = Quantity_Available,
                      fill = Product_Name)) +
```

```
  geom_bar(stat = "identity") +
  labs(
    title = "Quantity of Products Within Categories",
    x = "Product Category",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

Output:



3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

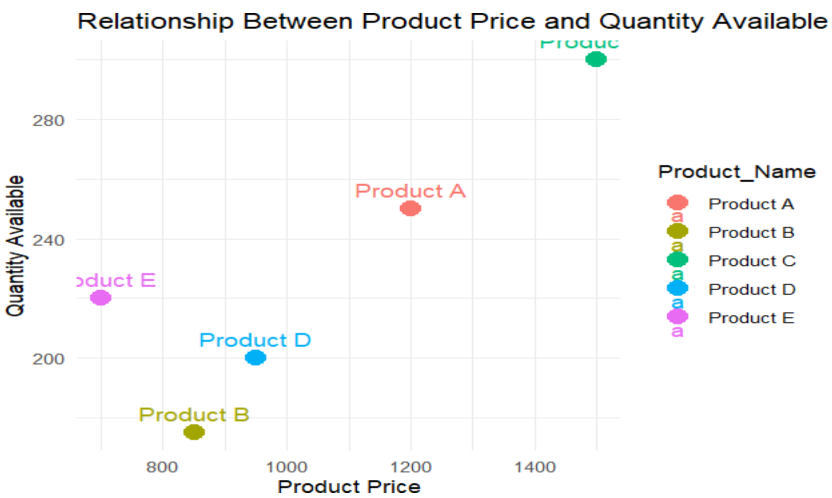
Program:

```
# Load library
library(ggplot2)


# Create dataset
inventory <- data.frame(
  Product_ID = c(1, 2, 3, 4, 5),
  Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Price = c(1200, 850, 1500, 950, 700),
  Quantity_Available = c(250, 175, 300, 200, 220)
)
# Scatter Plot
ggplot(inventory, aes(x = Price,
```

```
                    y = Quantity_Available,
                    color = Product_Name)) +
  geom_point(size = 4) +
  geom_text(aes(label = Product_Name),
            vjust = -0.8,
            size = 4) +
  labs(
    title = "Relationship Between Product Price and Quantity Available",
    x = "Product Price",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

Output:



Set-05: Website Analytics

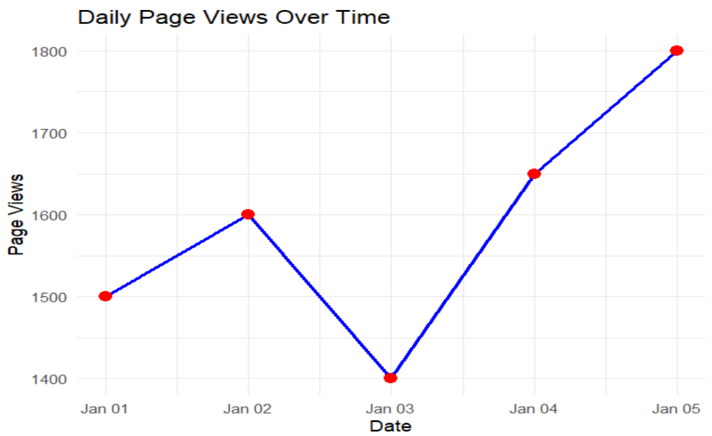
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Program:

```
# Load library
library(ggplot2)
# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
```

```
                "2023-01-02",
                "2023-01-03",
                "2023-01-04",
                "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
# Line Chart
ggplot(traffic, aes(x = Date, y = Page_Views, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views Over Time",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

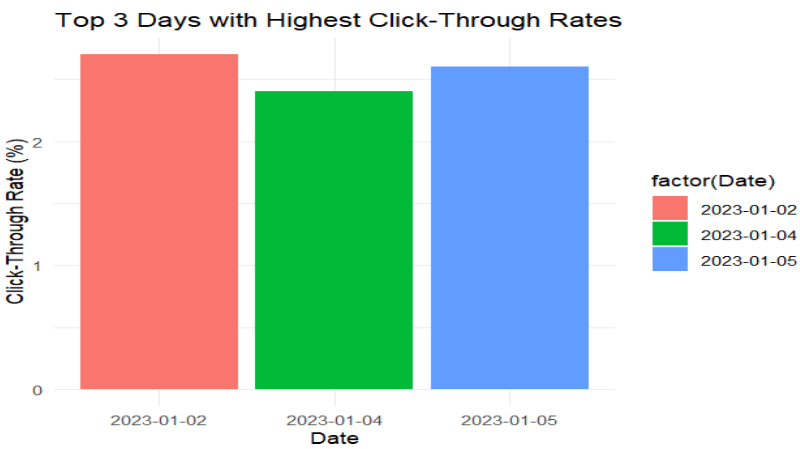
Program:

```
# Load library
library(ggplot2)# Create dataset
```

```r
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03",
                  "2023-01-04",
                  "2023-01-05")),
  Click_Through_Rate = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
# Select Top 3 Days
top_days <- traffic[order(-traffic$Click_Through_Rate), ][1:3, ]

# Bar Chart
ggplot(top_days, aes(x = factor(Date),
                     y = Click_Through_Rate,
                     fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top 3 Days with Highest Click-Through Rates",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```

Output:



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Program:

```
# Load libraries
library(ggplot2)
library(tidyr)
# Create dataset
interactions <- data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03",
                   "2023-01-04",
                   "2023-01-05")),
  Likes = c(120, 140, 110, 150, 170),
  Shares = c(40, 50, 35, 45, 55),
  Comments = c(30, 35, 25, 40, 45)
)
# Convert to long format
interaction_long <- pivot_longer(
interactions,
  cols = c(Likes, Shares, Comments),
  names_to = "Interaction",
  values_to = "Count"
)

# Stacked Area Chart
ggplot(interaction_long,
       aes(x = Date,
           y = Count,
           fill = Interaction)) +
  geom_area() +
```

```
labs(
  title = "Distribution of User Interactions",
  x = "Date",
  y = "Number of Interactions"
) +
theme_minimal()
```

Output: